



Original article

Molecular epidemiology, antibiogram profile and risk factor analysis of pathogenic *Escherichia coli* associated with pre-weaning diarrhoea in piglets from Haryana, India



Aman Deep, Kushal Grakh, Vaishali, Manesh Kumar, Renu Gupta, Dinesh Mittal, Ramesh Kumar, Naresh Jindal *

Department of Veterinary Public Health and Epidemiology, Lala Lajpat Rai University of Veterinary and Animal Sciences, Hisar 125004, India

ARTICLE INFO

Article history:

Received 18 April 2023

Received in revised form 29 June 2023

Accepted 23 August 2023

Keywords:

ETEC

India

Piglet diarrhoea

Risk factors

STEC

ABSTRACT

Background: Piglet diarrhoea is a multifactorial disease with serious implications for the swine industry worldwide, including India. The *Escherichia coli* (*E. coli*) pathotypes, i.e., enterotoxigenic *E. coli* (ETEC) and Shiga toxin-producing *E. coli* (STEC) are among the major bacterial agents attributed as causative agent for piglet diarrhoea, but studies related to genetic diversity, antibiogram profile and their correlation with risk factors of these pathogens are sparse.

Material and methods: A total of 104 faecal swab samples were collected from 32 different piggery units of Haryana, India and confirmed as *E. coli* by standard microbiological methods. The identified *E. coli* were characterized as ETEC and/or STEC using PCR assays and were studied for their genetic diversity by phylogenetic analysis of the sequences. All the isolates were subjected to antimicrobial susceptibility testing. Further, the correlation of variables with presence or absence of ETEC and/or STEC was also investigated by using Fisher's exact test.

Results: Microbiological isolation led to identification of 208 *E. coli* isolates. A total of 17.3% (31/208) isolates were characterized as ETEC and 4.8% (10/208) isolates as STEC, whereas 2.4% (5/208) isolates exhibited both ETEC and STEC pathotype. Of the total studied piggery units (n = 32), ETEC were isolated from fourteen and both ETEC and STEC from eight farms. The phylogenetic analysis of *Stx2* gene revealed 100% homology with *Stx2eA* variant from Germany, while analysis of *STII* gene revealed a distinct nucleotide and amino acid substitution when compared with standard strains. The antibiotic susceptibility testing revealed maximum resistance to moxifloxacin (71.9%) followed by tetracycline (58.1%) and amoxicillin with a total of 41.8% (87/208) *E. coli* isolates designated as multi-drug resistant (MDR). The multiple antibiotic resistance index varied from 0.05 to 0.75. The statistical analysis suggested three factors viz., number of farm worker(s), frequency of using disinfectant for floor cleaning and use of antibiotic in feed as risk factors significantly associated ($p < 0.05$) with ETEC associated diarrhoea at piggeries under study.

Conclusion: Current study warrants a need for systematic studies on the ETEC/STEC associated diarrhea and antibiotic resistance among these isolates to understand the mechanisms of origin and dissemination of drug resistant pathogens and to design suitable prevention and control measures to curb emergence of antibiotic resistance in the farm settings.

© 2023 The Authors. Published by Elsevier Ltd on behalf of King Saud Bin Abdulaziz University for Health Sciences. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

* Correspondence to: Lala Lajpat Rai University of Veterinary and Animal Sciences, Hisar, Haryana 125004, India.
E-mail address: nareshjindal1@gmail.com (N. Jindal).

Introduction

Piglet diarrhoea is one of the most frequently observed condition of the porcine industry worldwide including India. Diarrhoea in piglets is multifactorial as it involves interaction between sow, piglet, environment, etiological agents (virus, bacteria and parasites) and managerial practices [1]. Piglet diarrhoea in pre- (including neonatal) and post-weaning pigs due to pathogenic *E. coli* (colibacillosis) remains an economically important disease, which affects piglets during initial few weeks and post-weaning and generally characterised by sudden death or diarrhoea, stunted growth and dehydration [2,3]. Among *E. coli* pathotypes, Enterotoxigenic *E. coli* (ETEC) is the leading cause of the colibacillosis followed by less frequently isolated Shigatoxigenic *E. coli* (STEC) and Enteropathogenic *E. coli* (EPEC) [4]. The ETEC strains excrete heat-stable (STI and STII) and heat-labile (LT) toxins [5,6], whereas STEC strains produce Shiga-toxins (Stx1 and Stx2), which inhibit protein synthesis of host cells, thus leading to cell death [7]. Various risk factors such as stress, overcrowding, adverse climate and inefficient management practices, poor ventilation, type of feed and less frequent application of disinfectants in the piglet sheds may predispose the piglets to diarrhoea [8,9].

In order to control pre-weaning piglet diarrhoea, antibiotics are used non-judicially, which contribute to the emergence of resistant *E. coli* [10,11]. The antibiotic resistance further complicates the treatment and adds to economic losses in terms of cost incurred and mortality. The antibiotic resistant strains of *E. coli* along with plasmid harbouring antibiotic resistance genes can be easily disseminated to humans and other animals possibly through food chain [12,13]. Although a few studies are available avowing the prevalence, pathotyping, and risk factors for the occurrence of *E. coli* in post-weaned diarrheic piglets in India [14,15], but there is paucity of such studies on pre-weaned piglet diarrhoea. The present study was envisaged with the primary objective to isolate and characterize the *E. coli* pathotype(s) associated with pre-weaning diarrhea in piglets in the region, their genetic diversity and antibiotic susceptibility pattern. The second objective was to identify the risk factors associated with pre-weaning piglet diarrhoea or ETEC/STEC occurrence at the farms.

Materials and methods

Study design

The present cross-sectional study was carried out at different piggery units in the state of Haryana, India. The districts in the state under study were selected based on the pig population and the farms from each district were selected randomly. The study was approved by Institutional Animal Ethics Committee, and CPCSEA, New Delhi (1669/GO/REBiT-S/Re-L/12/CPCSEA). The required consents were obtained from the pig farmers to participate in the study and it did not involve any endangered/protected species. A total of 104 pre-weaned diarrhoeic piglets were included in this study. A comprehensive review of literature was conducted to develop a questionnaire comprising of 22 parameters pertaining to health of pigs and farm management practices (File-S1) for identification of potential risk factors for piglet diarrhoea (9,11,15). Pig farming in the state is done mostly by small and marginal farmers with education up to secondary level. Initially we developed a questionnaire (close and open-ended questions) and conducted a pilot study to determine whether piggery farmers are able to understand and answer the questions included with an aim to extract maximum information. The irrelevant and confusing questions, revealed by pilot study were removed and not included in the final questionnaire.

Sample and data collection

The sample size was calculated using the “Raosoft calculator” (<http://www.raosoft.com/samplesize.html?nosurvey>). The sample size of 98 was estimated based on a 18% prevalence of ETEC, a 10% margin of error, and a 99% confidence interval. The expected response proportion of 18% was assumed based on the studies conducted previously (9,15). A total of 104 rectal swab samples of diarrheic piglets aged between 2 days and 4 weeks were collected from 32 piggery units during March 2021 and October 2021. The samples were collected from eight districts of Haryana, India. An animal was considered to be suffering from diarrhoea if there was frequent passing of loose, watery and/or unformed faeces. One sample per affected animal was collected, with a maximum of five samples from each farm. The samples were transported on ice and processed on the same day in the laboratory. The epidemiological data was collected by face-to-face interview with the farm owners. The real time coordinates for the sampled farms were captured using Epicollect5 (<https://five.epicollect.net/>) to plot the precise location on the map with QGIS 3.22.7.

Risk factor analysis

A questionnaire was designed with 22 different questions on various factors hypothesized to be associated with the occurrence of Enterotoxigenic *E. coli* (ETEC) and Shiga toxin-producing *E. coli* (STEC). A pilot study was conducted by gathering the information from the farmers visiting the department for disease diagnosis of affected animals. The questionnaire was modified accordingly and then finalised by authors. The epidemiological data were recorded for further analysis in Excel spreadsheets. The data was coded for statistical analysis using SPSS software (ver. 4.2.0).

Isolation and characterization of *E. coli* pathotypes

The swab samples were plated directly on MacConkey agar (MLA) and incubated at 37°C for 24 h. Subsequently, two typical colonies were selected from the MLA plates, and were confirmed as *E. coli* based on the green metallic sheen on Eosin Methylene Blue (EMB) agar, Gram staining and biochemical (IMViC) test results. The colonies confirmed were provided isolate number accordingly. The genomic DNA extraction from *E. coli* isolates was carried out using a heat lysis/snap-chill method [16]. Briefly, 2–3 pure bacterial colonies were suspended in 150 µl nuclease free water, and placed in hot dry bath for 10 min and immediately snap-chilled on ice for 5 min. After centrifugation at 8000g, the supernatant was taken and stored as DNA for further use at –20 °C. All the oligonucleotides used in the study were got synthesized from Integrated DNA Technologies (USA) and are listed in Table 1. The extracted DNA was screened for *E. coli* genes encoding for the toxins (LT, STII and Stx2) using PCR, employing defined protocols. In brief, a reaction of 25 µl was used for PCR using the following conditions: 94 °C for 5 min; 94 °C for 45 s, 56 °C (for LT and Stx2)/60 °C (for STII) for 30 s, 72 °C for 45 s, total 35 cycles; and a step of final extension at 72 °C for 7 min. The amplified PCR products were analysed by agarose gel electrophoresis at a constant 90 V, using 2% agarose in TRIS acetate EDTA buffer (TAE) stained with 0.5 µl/ml of ethidium bromide. Then, the amplified products were visualized using UV transillumination in a gel doc system.

Sequencing and phylogenetic analysis

The PCR products for Stx2 and STII gene were purified using Gel extraction kit (Qiagen, Germany) and then got custom sequenced in both directions (AgriGenome Labs Pvt. Ltd., India). The sequences

Table 1
Oligonucleotides used for identification of *LT*, *STII* and *Stx2*.

Sr. No.	Description	Reference	Gene	Primers (5'-3')	Product size (bp)
1	Enterotoxin	Vidal et al.[17]	<i>LT</i>	F-GCGTCATCGTATACACAGGAGC R-GCACACGGAGCTCCTCAGTC	218
2	Enterotoxin	Vidal et al.[17]	<i>STII</i>	F-AATGTCCGTCTTGGCTTAGGAC R-AAAGGAGAGCTTCGTCACATTTT	421
3	Shiga toxin	Blanco et al.[18]	<i>Stx2</i>	F-CTTCGGTATCCTATTCCCGG R-CTGCTGTGACAGTGACAAAACGC	516

obtained were subjected to BLAST analysis to confirm the presence of gene identity and then submitted to GenBank for acquisition of accession numbers. The nucleotide (nt) sequences of both genes were aligned with other sequences retrieved from NCBI database using Clustal W algorithm in MEGA 11 programme [19]. The evolutionary history was inferred by constructing a phylogenetic tree using the Maximum Likelihood method and Kimura-2 (K2 +G) and Tamura-3 (T92) models [20]. The open reading frame (ORF) was translated using ORF finder (accessed at <https://www.ncbi.nlm.nih.gov/orffinder/>) and the dot plots of translated proteins for *STII* was plotted using BioEdit ver. 7.0.5.3 [21].

Antibiotic susceptibility testing

Antibiotic susceptibility testing was carried out by following the Clinical Laboratory Standards Institute guidelines (CLSI) and using the Kirby-Bauer disk diffusion method [22]. The following 20 antibiotics belonging to seven different classes were employed for the susceptibility testing; amikacin (AK, 30 µg), amoxicillin/clavulanic acid (AMC, 30 µg), ampicillin (AMP, 10 µg), cefadroxil (CFR, 30 µg), cefatoxime/clavulanic acid (CEC, 30 µg), cephalixin (CN, 30 µg), cefepime (CPM, 30 µg), ceftriaxone (CTR, 10 µg), co-trimoxazole (COT, 25 µg), chloramphenicol (C, 30 µg), doxycycline (DO, 30 µg), enrofloxacin (EX, 10 µg), gentamicin (GEN, 10 µg), kanamycin (K, 30 µg), levofloxacin (LE, 5 µg), moxifloxacin (MO, 5 µg), nalidixic acid (30 µg), streptomycin (S, 10 µg), tetracycline (TE, 10 µg) and neomycin (N, 30 µg). The ATCC 25922 present in the laboratory was used as quality control strain. The results were interpreted as per the CLSI guidelines. Similar to Grakh et al. [23] for the purpose of current study, the isolates with intermediate susceptibility were taken as not resistant and isolates were considered multiple drug resistance (MDR) if they were resistant to three or more antibiotic classes. Multiple antibiotic resistance (MAR) index was calculated as the ratio between the number of antibiotics that an isolate is resistant to and total number of antibiotics to which the test isolate has been evaluated for susceptibility [24].

Statistical analysis

All the collected questionnaire data was analysed by Fisher's exact test using SPSS software. The Fisher's exact test was used to analyse any association of various risk factors with occurrence of ETEC and STEC associated diarrhoea at the farm. The data related to virulence gene frequency in *E. coli* isolates and antibiotic resistance were analysed using chi-square test. A value of $p < 0.05$ was considered as significant for associations between variables (risk factors).

Cluster analysis

A clustered heat map (double dendrogram) was constructed to depict the presence of *LT*, *STII* and *Stx2* genes and MDR status of different pathotypes among *E. coli* isolates used in the current study. Data were processed in a binary matrix using the NCSS (trial version)

(NCSS, LLC, USA) software package. The isolates were grouped using agglomeration method, based on the unweighted average distance.

Results

Isolation and characterization of *E. coli* pathotypes

Based on the characteristic colony on MLA and EMB, Gram staining, and test results of IMViC, 208 isolates were confirmed as *E. coli* from 104 samples processed. All the rectal swab samples (100%) were found to be positive for *E. coli* by using standard microbiological methods. These samples were used for further study. Based on the results of PCR, 7.21% (15/208) *E. coli* isolates harboured only *LT* gene, 7.21% (15/208) isolates harboured only *STII* gene and one isolate (0.48%) was positive for both *LT* and *STII* genes suggesting a total of 31 (14.9%) isolates to be ETEC pathotype. The *Stx2* gene specific to STEC pathotype specific product was amplified among 10 (4.8%) *E. coli* isolates. Interestingly, five isolates showed both ETEC and STEC pathotype evident from their amplification of both pathotype specific amplicons. Based on number of animals tested, 27/104 (25.7%) animals were positive for ETEC pathotype and 10/104 (9.6%) for STEC pathotype, whereas four animals (3.8%) were positive for both ETEC/STEC pathotypes. Overall, 41 (39.42%) piglets showed the presence of either of the pathotype specific markers. The results on the isolate as well as animal basis have been depicted in Table 2.

Sequencing and phylogenetic analysis

The five STEC isolates sequenced for *Stx2* gene received accession numbers as; OQ627399 (ECO441), OQ627400 (ECO1621), OQ627401 (ECO1821), OQ627402 (ECO2121) and OQ627403 (ECO3012). All these sequences were aligned with previously published sequences retrieved from GenBank and revealed a maximum of 100% nucleotide (nt) identity with *Stx2eA* variant of *Stx2* (Acc. No. FM998844.1) from Germany and a minimum of 90.7% homology with *Stx2j* variant (MZ571121.1) (Fig. 1). Three ETEC isolates sequenced for *STII* gene were accessioned as OQ615950 (ECO1231), OQ615951 (ECO211) and OQ615952 (ECO911). On aligning with retrieved sequences these isolates revealed more than 99% nt homology with previously published *STII* gene as shown in Fig. 2. The conservation plot of translated ORF of *STII* gene with six *E. coli* strains retrieved from NCBI BLAST revealed substitution of histidine with asparagine at 35th amino acid position, which corresponds to 103rd nucleotide change within ORF of the *STII* gene. Fig. 3.

Antibiotic susceptibility testing

A total of 208 *E. coli* isolates were tested for antibiotic susceptibility against 20 antibiotics and the results showed maximum resistance for moxifloxacin (71.9%), followed by tetracycline (58.1%), amoxicillin (44.3%) and for ampicillin (40.9%). The rest of the antibiotics showed less than 40% resistance among *E. coli* isolates as depicted in the Fig. 4. On contrary, all the tested isolates were 100% sensitive to gentamicin followed by cefotaxime (99%), amikacin (97.5%), chloramphenicol (94.1%), cephalixin and cefepime (85.7%),

Table 2

Presence of virulence markers of ETEC and STEC.

Group	LT alone	STII alone	LT and STII	Stx2 alone	LT and Stx2	STII and Stx2
<i>E. coli</i> isolates (n = 208)	15 (7.2%)	15 (7.2%)	01 (0.4%)	10 (4.8%)	03 (1.4%)	02 (0.9%)
Number of animals (n = 104)	14 (13.4%)	12 (11.5%)	01 (0.96%)	10 (9.6%)	03 (2.9%)	01 (0.96%)

levofloxacin (81.8%), cefadroxil and ceftriaxone (77.3% each), streptomycin (72.9%), and kanamycin, neomycin and enrofloxacin (72.4% each). A total of 87 (41.8%) isolates were MDR across different districts of Haryana as depicted in Fig. 5. MAR Index was calculated for individual isolate and it was found that 28.4% of the current study isolates had MAR index of more than 0.3. The MAR index varied from 0.05 to 0.75. Presence of LT gene with amikacin resistance and the presence of Stx2 gene with nalidixic acid, enrofloxacin and levofloxacin resistance were evident among the isolates ($p < 0.05$).

Cluster analysis

A cluster analysis was performed to group pathotype into different clusters based on tested genes and MDR status of the isolates. A total of three different clusters were obtained. All the STEC and mixed (ETEC/STEC) pathotypes were clustered in cluster A while all the ETEC were clustered equally among clusters B and C (Fig. 6).

Risk factor analysis

The statistical analysis of data collected using questionnaire is depicted in Table 3. On analysis, ETEC associated diarrhoea was found to be significantly associated ($p < 0.05$) with three factors/variables; i.e., number of farm worker[s], frequency of using

disinfectant for floor cleaning and use of antibiotic in feed (Table 4). The farms (n = 25) engaging one to three farm workers showed higher cases of ETEC associated diarrhoea in the animals housed as compared to farms with more than three persons as farm labour. However, more ETEC associated diarrhoea was observed with some other factors such as more farms in the neighbour, group housing rearing system and self-processed feed, but significant association for these factors was not established. Odds of having 1–3 farm workers at farms was found to be 10 times higher on the farms which were ETEC positive as compared to farms without ETEC. The variables, i.e., number of farm workers, frequency of using disinfectant for floor cleaning, and use of antibiotic in feed were not found to be associated significantly with respect to occurrence of STEC. No significant association for any of the study variables was observed with STEC.

Discussion

We conducted this study to determine the pathotypes of *E. coli* involved in piglet diarrhoea in pre-weaned piglets within selected piggery units in different districts of the Haryana state. In addition, the study identified the antimicrobial resistance pattern of the isolates and the risk factors associated with the occurrence of pathotypes with the piglet diarrhoea. Piglet diarrhoea is an important condition from economic view point, which is usually associated with high mortality resulting in decline of the pig population, increased cost of treatment, and losses due to condemnation of carcass [24]. In India, piglet mortality resulting from diarrhoea has been

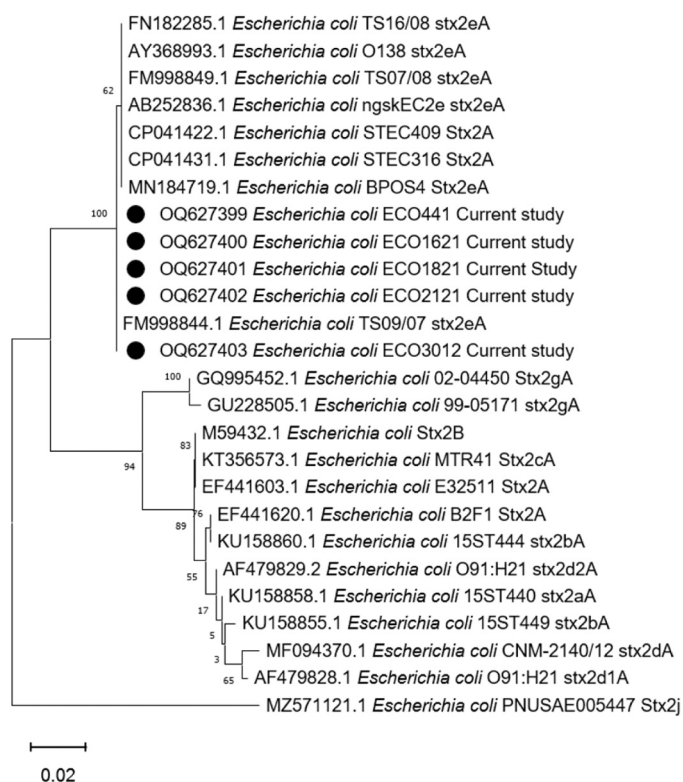


Fig. 1. Phylogenetic analysis based on partial nucleotide sequence of Stx2 gene. The phylogenetic relationship was constructed using the Maximum Likelihood method with Kimura-2 parameter (K2+G) model in MEGA 11 software. Texas exhibiting GenBank accession numbers and variant of Stx2 gene were used for comparison. Bootstrap values (%) are represented at each tree node. Node support was assessed with 500 bootstrap pseudo-replicates. The current study isolates are shown with black dots.

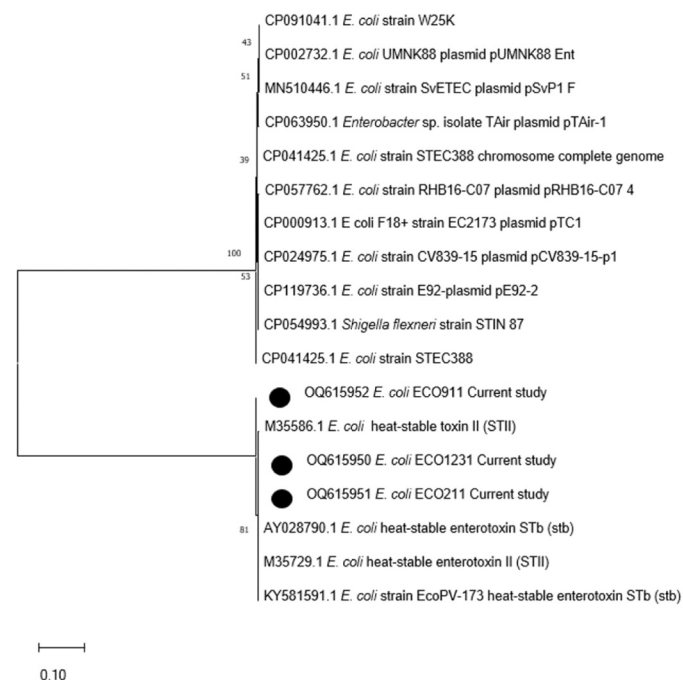


Fig. 2. Phylogenetic analysis based on partial nucleotide sequence of STII gene. The phylogenetic relationship was constructed using Maximum Likelihood method with Tamura-3 parameter (T92) model in MEGA 11 software. Texas exhibit GenBank accession numbers of strains used for comparison. Bootstrap values (%) are represented at each tree node. Node support was assessed with 500 bootstrap pseudo-replicates. The current study isolates are shown with black dots.

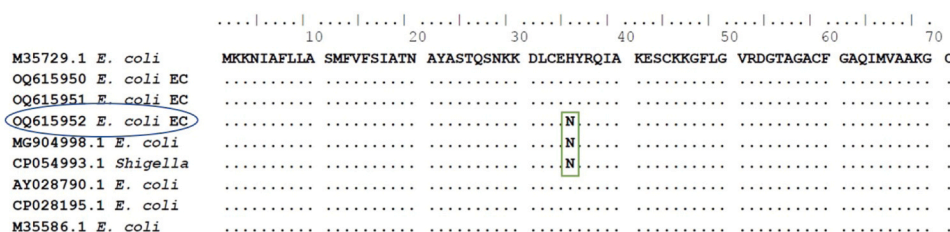


Fig. 3. Dot plot of translated ORF of STII gene of *E. coli* isolates. *E. coli* (M35729.1) was used as standard strain followed by three current study isolates viz., OQ615950, OQ615951, OQ615952 and sequences retrieved from NCBI. The substitution at 35th amino acid position is marked in green box and current study isolate showing substitution is marked in oval shape.

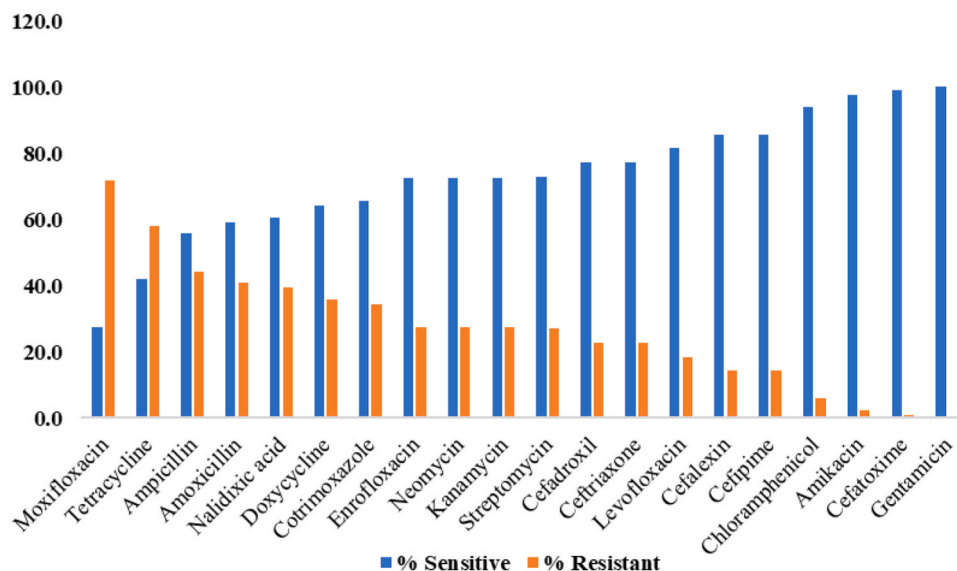


Fig. 4. Antimicrobial susceptibility pattern of *E. coli* isolates from diarrheic piglets.

reported to be the top most contributing factors to economic loss in the piggery industry [25,26]. Piglet diarrhoea is multifactorial in nature and the involvement of pathogen may vary with region. *E. coli* and *Salmonella* are the major etiological agents involved in piglet diarrhoea globally [27]. *E. coli* particularly the ETEC [26] and less commonly the STEC pathotype [28], are important determinants of piglet diarrhoea affecting both pre- and post-weaning piglets.

The current study reported a total of 14.9% (31/208) isolates as ETEC by amplifying either or both of pathotype specific markers viz., *LT* and or *STII* genes. The observations of present study are in consensus with the findings of Shakuntala et al. [15], wherein authors reported 15.1% *E. coli* isolates as ETEC. However, higher percentage of *E. coli* isolates (29.93% and 67.8%) were characterized as ETEC by Rajkhowa et al. [29] in India and Do et al. [30] in Korea, respectively. The variation in the occurrence of ETEC may be due to different geographical locations, which is a subject of investigation to understand the epidemiological factors influencing its occurrence. Sanitation and hygiene of water and the associated environment plays a key role in control and prevention of the diarrheal disease, as investigated in humans [31]. In the present study, 48.38% ETEC isolates harboured *LT* and *STII* gene each. Rajkhowa et al. [29] also reported *LT* and *STII* genes among 40.4% and 48.9% of ETEC isolates, respectively. In contrast, Zhang et al. [32] and Kim et al. [33] reported lower prevalence of 18–19% and 23–25% for *LT* and *STII*, respectively.

Based on the amplification of *Stx2* marker, a total of 7.2% isolates were characterized as STEC. The studies conducted by Kylla et al. [34], Rajkhowa and Sarma [35], Kaufmann et al. [36] and Shakuntala et al. [15] also reported the presence of *Stx2* marker among STEC isolates. Globally, several studies have reported varying prevalence

of STEC from samples of pigs with a tune of 30.2%, 4.5% and 0.4% from Germany, Argentina and South Africa, respectively [37–39]. The variations observed in different studies with regard to prevalence of ETEC and STEC might be accredited to the sample size, age, breed, health status of animals, diverse climatic conditions, dissimilar animal husbandry practices in different geographical locations of these farms as has been reported for several other *E. coli* pathotype of diverse origin [23]. Additionally, different sampling techniques, isolation technique, and/or the criterion to define STEC may also contribute to variations in prevalence across studies.

All the five STEC isolates of the current study showed 100% homology with *Stx2eA* variant of *Stx2* (Acc. No. FM998844.1) from Germany, suggesting that strains harbouring *Stx2eA* variant are frequent in the region of the study. Additionally, it could be inferred that isolates might have originated from a common source and thus highlights the importance of global surveillance and sharing of sequence data for timely management of *E. coli* infections.

Two ETEC isolates revealed (ECO1231 and ECO211) 100% homology with ETEC isolates from retrieved sequences, whereas both these isolates showed 99.50% homology with the third ETEC isolate of current study (ECO911). It might be attributed to a single nucleotide mutation at 103th position in the ORF region of *STII* gene as depicted in the Fig. 4, which resulted in the change of an amino acid at 35th position i.e., asparagine to histidine. Histidine and asparagine are both amino acids with different chemical properties, and their substitution could potentially alter the structure and function of the protein encoded by the *STII* gene. This could affect the virulence or pathogenicity of the *E. coli* strain, and possibly impact its ability to cause disease [40]. Further studies would be

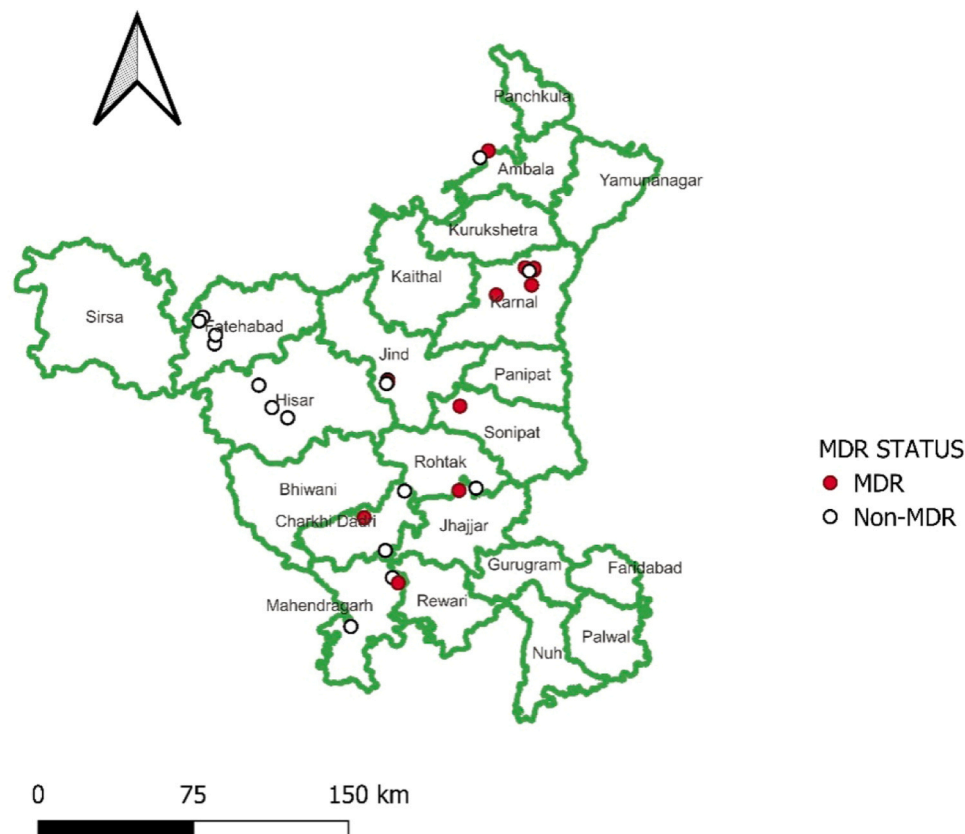


Fig. 5. The distribution of multidrug resistant *E. coli* in piggery units of different districts in state of Haryana, India.

required to investigate the functional consequences of this amino acid substitution and its potential role in the virulence of the *E. coli* strain. As expected, ECO911 revealed maximum homology with *E. coli* isolates (CP068040.1 and MG904998.1), with same substitution of nucleotide and amino acid at respective positions. These findings suggests that the STEC and ETEC isolates found in the study population are closely related to previously published strains, with some genetic diversity observed among the ETEC isolates. The identification of these strains and their genetic diversity is important for understanding their epidemiology and potential for transmission to humans.

All the antibiotics were found to have varied degree of resistance except gentamicin for which all the isolates were sensitive. Even though enrofloxacin was reported to be the most commonly used antibiotic at the farms under study but maximum resistance recorded against moxifloxacin, a fluoroquinolone (71.9%) is surprising. As none of the farmers reported the use of moxifloxacin, it needs further investigations to arrive at the reasons of higher resistance to the aforesaid antibiotic. Moxifloxacin, however, nowadays is widely used in the region for therapeutic treatment of large and small ruminants, which might have led to environmental contamination and thus transfer to swine population [41]. The resistant *E. coli* isolates might have easily disseminated to young piglets from various farm components, equipment and other environmental sources (42). The overall resistance of quinolones varied from 18.3% to 72.6%. Higher resistance to nalidixic acid among swine origin *E. coli* isolates has been reported earlier by Rajkhowa et al. [29] and Rajkhowa and Sarma [35], and Yang et al. [41]. This may be attributed to varying degree of fluoroquinolones use in various geographical regions. Moreover, by the questionnaire findings, it was observed that in this part of country the pig farmers rarely use the fluoroquinolones for therapeutics in various affections of pigs, except enrofloxacin. The resistance to fluoroquinolones thus raises public health concerns as

it is widely used in human medicine for treatment. The resistance to second most commonly used antibiotic (as reported in current study) tetracycline (58.1%) may be attributed to its overuse or non-judicious application for the treatment purpose by the farmers. Although resistance to other antibiotics which are rarely used at pig farms such as amoxicillin, ampicillin, nalidixic acid and doxycycline are of great concerns as these antibiotics are also used for the treatment of human ailments and their possible dissemination to human population can make the treatment a daunting task. The resistance to these antibiotics might have developed by plasmid exchange from other resistant pathotype of *E. coli* present in the environment [42,43]. Additionally, movement of labour or visitors' and animals during purchase of the pigs might be attributed as source of exchange of resistant bacteria across farms. All the *E. coli* isolates were, however, sensitive to gentamicin, which suggests that the antibiotic might still have not been used at piggery units in Haryana. The similar finding was also observed by Grakh et al. [23] with regards to gentamicin on commercial broiler poultry farms. Another concern raised in the study is high number of MDR *E. coli* isolates as it limits the choice of antimicrobials available for treatment of bacterial infections. These findings might help veterinarians to choose the most effective antibiotic for treating diarrhoea in piglets, with an understanding of the current antimicrobial resistance patterns at local level. The cluster analysis provided additional insights into the genetic relatedness and antibiotic resistance patterns of the isolates. The grouping of the pathotypes into different clusters based on the tested genes and MDR status of the isolates may have important implications for the development of effective prevention and control measures.

In the present study, farms with a smaller number of farm workers had significantly higher incidence of ETEC associated diarrhoea as compared to farms with a greater number of workers ($p < 0.05$). Grakh et al. [44] had reported the higher incidence, of

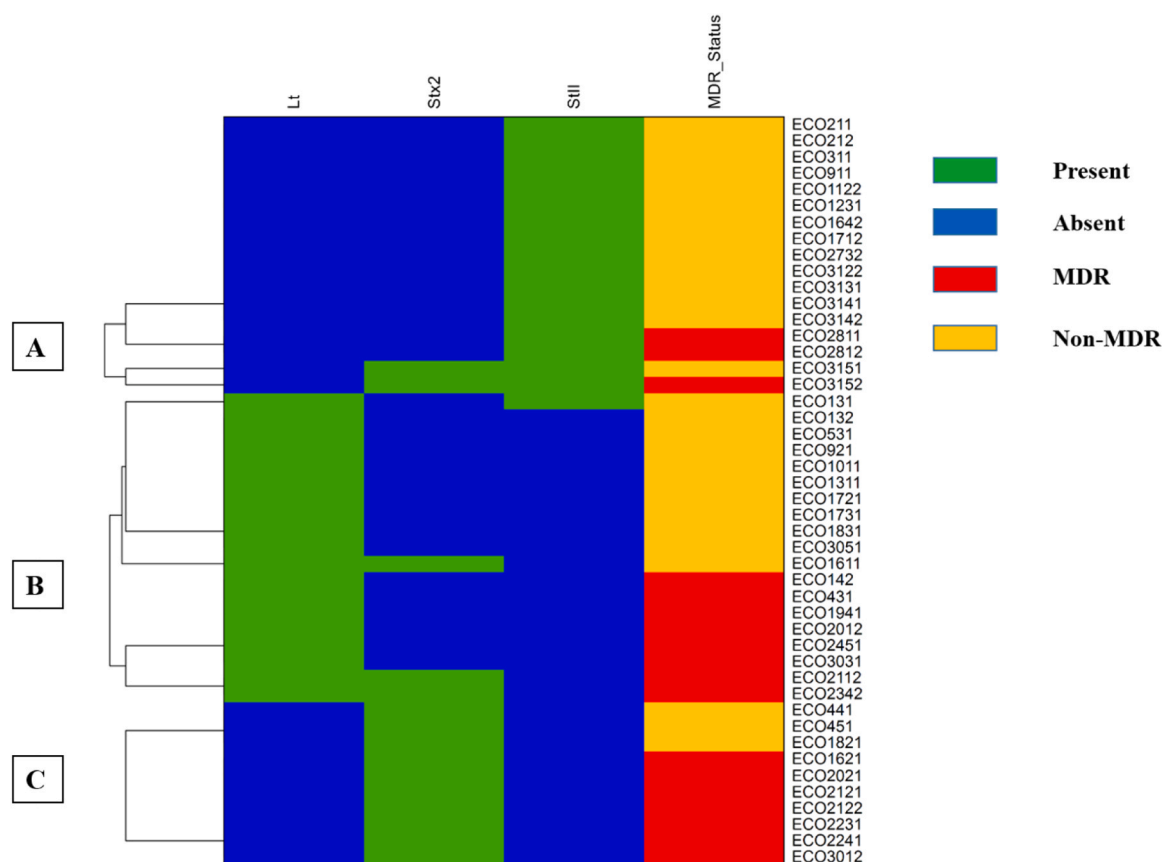


Fig. 6. Clustered heat map (double dendrogram) depicting the distribution of *LT*, *STII* and *Stx2* gene and MDR status among pathogenic *E. coli* isolates (ETEC, STEC and ETEC/STEC). The leftmost portion of this figure is the dendrogram resulting from cluster analysis. There appear three clusters of isolates: (A), (B) and (C). Cluster A contains all the STEC and mixed (ETEC/STEC) pathotypes, cluster B and C contained all the ETEC isolates with nearly equal distribution based on their MDR status. Just to the right of the dendrogram are columns 1–4 which shows the virulence genotype of isolates tested. Each column in this group shows the results for a single gene or trait. The identity of each gene is shown at the top of the diagram. Column five shows the isolates number/ids.

avian colibacillosis on the poultry farms with less farm workers, as the same worker has to perform multiple operations leading to frequent contact with infected and apparently healthy piglets. These workers might be mechanically disseminating the infectious agents between infected and non-infected farms, owing to multiple time movement between different sections. The other possible reason for the finding might be attributed to the improper management of litter and hygiene of the farm due to lack of labour. The less frequent application of disinfectant at farms for cleaning was significantly associated with higher incidences of ETEC associated diarrhoea in the present study. The similar findings were observed by the study conducted by VinodhKumar et al. [9]. The frequent application of disinfectants on the farm reduces the bacterial load and prevents farm from infections [45].

Increased use of antimicrobials in feed at the piggery farms was found to be significantly associated with ETEC occurrence, although this finding is surprising. It could be assumed that for preventing diarrhoea at piggery farms, the farmers might be using antimicrobials and that might have attributed to the resistance in these farms. The other reason might be the assumption of farmers that antibiotics will provide protection against everything leading to undermining the importance of biosecurity. The similar findings also had been reported by Grakh et al. [44], wherein incidence of *E. coli* infection was higher on poultry farms with frequent use of antibiotics. However, further studies are warranted to correlate the ETEC or STEC with use of antimicrobials over a period of time. Piglets with less frequency of diarrhoea per day revealed higher prevalence of ETEC on the farms, which might be related to the pathogenic mechanism of the ETEC pathotype. However, the frequency of diarrhoea is a post

disease event and cannot be taken as risk factor, but it might be helpful to imply timely preventive or control interventions. Earlier studies have emphasized that inadequate management practices, lack of biosecurity measures, less frequent disinfection, use of shallow water, overcrowding, poor ventilation and high temperature are the important risk factors predisposing the piglets to diarrhoea [9,46]. On the other hand, the results showed that more ETEC associated diarrhea was observed with some other factors such as more farms in the neighbour, group housing rearing system, and self-processed feed, but significant association for these factors was not established. This may be due to the complex interplay of multiple factors involved in ETEC infections. These findings may help inform strategies to reduce the prevalence of ETEC infections in farm animals and improve animal health.

Interestingly, the study found no significant association for any of the study variables with STEC. This suggests that the risk factors for STEC may differ from those for ETEC, and further research is needed to identify the factors associated with STEC infections.

In the present study, *E. coli* associated with piglet diarrhoea, their antimicrobial susceptibility profile and genetic relatedness, and associated risk factors were investigated, which helped to understand the infection and factors associated. The study also improved the available knowledge and understanding in the field of epidemiology relating to association of *E. coli* with piglet diarrhoea.

However, there are certain limitations, which need to be addressed in further studies. We did not investigate other potential etiological agents of piglet diarrhoea, particularly viral and protozoal causes, which could have provided a more comprehensive understanding of the diarrheal etiology. Moreover, the study region was

Table 3

Risk factor analysed for possible association with ETEC/STEC associated diarrhoea.

Sr. No.	Variable	Category	Total farms tested	% ETEC positive farms	% STEC positive farms
1.	No. of farm in 5 km radius	1–3	22	59	18
2.	No. of sheds	4–6	10	90	40
		1–5	31	67	25
		> 5	1	10	0
3.	Total no. of animals	20–220	20	80	25
		220–420	7	40	28
		> 420	5	60	33
4.	Number of piglets	10–80	19	78	40
		80–150	7	71	28
		150–220	6	33	0
5.	Type of housing	Individual	3	60	66
		Group Housing	29	68	20
6.	Feeding	Concentrate	28	67	28
		Swill feeding	4	75	0
7.	Type of feed made	Self	8	87	25
		Readymade	24	62	25
8.	Hygiene	Good	16	75	31
		Normal	16	62	18
9.	Class of antibiotic used	Penicillin	5	80	0
		Cephalosporin	21	66	33
		Sulphonamide	6	66	16
10.	Water source	Shallow well	5	80	20
		Deep well	27	67	25
11.	How often pens are cleaned	Once a day	11	72	27
		Twice a day	21	67	23
12.	Ventilation system	Natural	30	73	26
		Exhaust	2	0	0
13.	Age (days)	0–10	1	0	0
		11–20	22	68	36
		21–30	9	77	0
14.	Clinical condition of piglet	Normal	27	66	29
		Weak	5	80	0
15.	Type of diarrhoea	Watery	20	65	25
		Semi solid	12	75	25
16.	Colour of faeces	Whitish	5	80	0
		Yellow	27	66	29

Table 4

Risk factors significantly associated with ETEC associated diarrhoea.

Sr. No.	Variable	Category	Total farms tested	Number of ETEC positive farms	% ETEC positive farms	p value
1.	Number of farm worker	1–3	25	20	80	0.019
		4–6	7	2	28	
2.	Use of antibiotic in feed	Yes	8	8	100	0.035
		No	24	14	58	
3.	Frequency of using disinfectant for floor cleaning	Daily	7	2	28	0.019
		Weekly	25	20	80	

confined to the state of Haryana, India, only. So, the similar studies with a greater number of samples and regions may be undertaken for better understanding the disease dynamics of these pathogens. Further, study in different regions will pave the way to determine the effects of various husbandry practices, environmental conditions, and antimicrobial usage patterns on the pathogens under study.

Conclusions

In conclusion, this study provides valuable insights into the prevalence, pathotypes, and antimicrobial resistance patterns of *E. coli* associated with piglet diarrhea. The findings suggest that ETEC and STEC are important pathogens involved in piglet diarrhea in the

region, and that there is a need of effective measures to control and prevent the disease. Future studies with focus on investigating the epidemiological factors influencing the occurrence of ETEC and STEC in this region, as well as on the functional consequences of the amino acid substitution identified in the *STII* gene of some of the isolates are needed. Additionally, the global surveillance and sharing of sequence data for effective management of *E. coli* infections could be further strengthened based on the insights from this study. Managing the risk factors associated with piglet diarrhoea might help to reduce the occurrence of this condition and thus help in reducing the economic losses. Further, effective vaccination programs in pig farms are need of the hour to curb this menace of antimicrobial resistance among such pathogenic *E. coli* isolates and thus to minimize the impact on animal, human and environmental health.

Financial Disclosures

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors also declare that they have no conflict of interests with authorship, publication and data utilized in the study. All the authors have agreed to submit the manuscript for possible publication evaluation with Journal of Infection and Public Health.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2023.08.019.

References

- [1] Hong TTT, Linh NQ, Ogle B, Lindberg JE. Survey on the prevalence of diarrhoea in pre-weaning piglets and on feeding systems as contributing risk factors in smallholdings in Central Vietnam. *Trop Anim Health Prod* 2006;38(5):397–405 (<https://link.springer.com/article/10.1007/s11250-006-4399-z>).
- [2] Fairbrother JM, Nadeau É, Gyles CL. *Escherichia coli* in postweaning diarrhea in pigs: an update on bacterial types, pathogenesis, and prevention strategies. *Anim Health Res Rev* 2005;6:17–39. <https://doi.org/10.1079/AHR2005105>
- [3] Xu G, An W, Wang H, Zhang X. Prevalence and characteristics of extended-spectrum β -lactamase genes in *Escherichia coli* isolated from piglets with post-weaning diarrhea in Heilongjiang province, China. *Front Microbiol* 2015;6:1103. <https://doi.org/10.3389/fmicb.2015.01103>
- [4] Luppi A. Swine enteric colibacillosis: Diagnosis, therapy and antimicrobial resistance. *Porc Health Manag* 2017;3. <https://doi.org/10.1186/S40813-017-0063-4>
- [5] Francis D.H. Diagnostic notes Non refereed Enterotoxigenic *Escherichia coli* infection in pigs and its diagnosis, vol. 10, 2002.
- [6] Fleckenstein JM, Hardwidge PR, Munson GP, Rasko DA, Sommerfelt H, Steinsland H. Molecular mechanisms of enterotoxigenic *Escherichia coli* infection. *Microbes Infect* 2010;12:89–98. <https://doi.org/10.1016/j.micinf.2009.10.002>
- [7] Melton-Celsa AR. Shiga Toxin (Stx) classification, structure, and function. *Microbiol Spectr* 2014;2. <https://doi.org/10.1128/MICROBIOLSPPEC.EHEC-0024-2013>
- [8] Laine TM, Lyytikäinen T, Yliäho M, Anttila M. Risk factors for post-weaning diarrhoea on piglet producing farms in Finland. *Acta Vet Scand* 2008;50. <https://doi.org/10.1186/1751-0147-50-21>
- [9] VinodhKumar OR, Singh BR, Sinha DK, Pruthivishree BS, Tamta S, Dubal ZB, et al. Risk factor analysis, antimicrobial resistance and pathotyping of *Escherichia coli* associated with pre- and post-weaning piglet diarrhoea in organised farms, India. *Epidemiol Infect* 2019;147. <https://doi.org/10.1017/S0950268819000591>
- [10] Nirupama KR, Vinodh VK, Pruthivishree BS, Sinha DK, Murugan MS, Krishnaswamy N, et al. Molecular characterisation of blaOXA-48 carbapenemase-, extended-spectrum β -lactamase- and Shiga toxin-producing *Escherichia coli* isolated from farm piglets in India. *J Glob Antimicrob Resist* 2018;13:201–5. <https://doi.org/10.1016/j.jgar.2018.01.007>
- [11] Mittal D, Grakh K, Prakash A, Moudgil P, Devi B, Jadhav V. Isolation, Identification and Characterization of Enteric Bacteria from Post Weaning Diarrheic Pigs and

- their Resistance to Multiple Antibiotics. *Int J Curr Microbiol Appl Sci* 2018;7:2377–84. <https://doi.org/10.20546/IJCMAS.2018.712.269>
- [12] Manges AR, Johnson JR. Food-borne origins of *Escherichia coli* causing extraintestinal infections. *Clin Infect Dis* 2012;55:712–9. <https://doi.org/10.1093/cid/cis502>
- [13] Grakh K, Mittal D, Prakash A, Moudgil P, Kumar P, Kumar R. Molecular Detection of Antimicrobial Resistant genes with special reference to ESBL in *E. coli* from Diarrheic Piglets. *Indian J Vet Sci Biotechnol* 2021;17:58–63. <https://doi.org/10.21887/IJVS BT.17.2.11>
- [14] Ribbens S, Dewulf J, Koenen F, Mintiens K, de Sadeleer L, de Kruif A, et al. A survey on biosecurity and management practices in Belgian pig herds. *Prev Vet Med* 2008;83:228–41. <https://doi.org/10.1016/j.PREVETMED.2007.07.009>
- [15] Shakuntala I, Sanjukta RK, Das S, Puro K, Ghatak S, Rajkhowa S, Milton AA, Sen A. Occurrence and characterization of *Escherichia coli* isolated from diarrhoeic piglets in Meghalaya. *Indian J Hill Farm* 2018 2017 Special Issue (<https://epubs.icar.org.in/index.php/IJHF/article/view/83988>).
- [16] Englen MD, Kelley LC. A rapid DNA isolation procedure for the identification of *Campylobacter jejuni* by the polymerase chain reaction. *Lett Appl Microbiol* 2000;31:421–6. <https://doi.org/10.1046/j.1365-2672.2000.00841.X>
- [17] Vidal R, Vidal M, Lagos R, Levine M, Prado V. Multiplex PCR for diagnosis of enteric infections associated with diarrheagenic *Escherichia coli*. *J Clin Microbiol* 2004;42:1787–9. <https://doi.org/10.1128/JCM.42.4.1787-1789.2004>
- [18] Blanco M, Blanco JE, Mora A, Dahbi G, Alonso MP, González EA, et al. Serotypes, virulence genes, and intimin types of Shiga toxin (verotoxin)-producing *Escherichia coli* isolates from cattle in Spain and identification of a new intimin variant gene (*eae-xi*). *J Clin Microbiol* 2004;42:645–51. <https://doi.org/10.1128/JCM.42.2.645-651.2004>
- [19] Tamura K, Stecher G, Kumar S. MEGA 11: Molecular Evolutionary Genetics Analysis Version 11. *Mol Biol Evol* 2021. <https://doi.org/10.1093/molbev/msab120>
- [20] Tamura K, Nei M. Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Mol Biol Evol* 1993;10(512–26):0023. <https://doi.org/10.1093/OXFORDJOURNALS.MOLBEV.A04>
- [21] Hall T.A. BioEdit: a user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT n.d., 1999.
- [22] Bauer AW, Kirby WM, Sherris JC, Turck M. Antibiotic susceptibility testing by a standardized single disk method. *Tech Bull Regis Med Technol* 1966;36:49–52.
- [23] Grakh K, Mittal D, Prakash A, Jindal N. Characterization and antimicrobial susceptibility of biofilm-producing Avian Pathogenic *Escherichia coli* from broiler chickens and their environment in India. *Vet Res Commun* 2022;46:537–48. <https://doi.org/10.1007/S11259-021-09881-5>
- [24] Kruperman PH. Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of fecal contamination of foods. *Applied. Environ Microbiol* 1983;46:165–70.
- [25] Roy R, Mondal T, Moktan MW. Patterns of pre-weaning piglet mortality and economic losses in field condition. *Afr J Agric Res* 2018;13:1291–6. <https://doi.org/10.5897/ajar2015.10090>
- [26] Sinha R, Sahoo NR, Shrivastava K, Kumar P, Qureshi S, De UK, et al. Resistance to ETEC F4/F18-mediated piglet diarrhoea: opening the gene black box. *Trop Anim Health Prod* 2019;51:1307–20. <https://doi.org/10.1007/S11250-019-01934-X>
- [27] Abonyi F, Omeh C, Machebe N. Neonatal mortality of pigs in Nsukka, Southeast Nigeria. *Afr J Biotechnol* 2016;11:13228–34. <https://doi.org/10.4314/ajb.v11i68>
- [28] Mandakini R, Dutta TK, Chingtham S, Roychoudhury P, Samanta I, Joardar SN, et al. ESBL-producing Shiga-toxigenic *E. coli* (STEC) associated with piglet diarrhoea in India. *Trop Anim Health Prod* 2015;47:377–81. <https://doi.org/10.1007/s11250-014-0731-1>
- [29] Rajkhowa S, Neher S, Pegu SR, Sarma DK. Bacterial diseases of pigs in India: A review. *Indian J Comp Microbiol Immunol Infect Dis* 2018;39:29. <https://doi.org/10.5958/0974-0147.2018.00014.4>
- [30] Do KH, Byun JW, Lee WK. Virulence genes and antimicrobial resistance of pathogenic *Escherichia coli* isolated from diarrheic weaned piglets in Korea. *J Anim Sci Technol* 2020;62:543. <https://doi.org/10.5187/JAST.2020.62.4.543>
- [31] Hutton G, Chase C. Water supply, sanitation, and hygiene, disease control priorities, third edition (Volume 7). *Inj Prev Environ Health* 2017:171–98. https://doi.org/10.1596/978-1-4648-0522-6_CH9
- [32] Zhang W, Zhao M, Ruesch L, Omot A, Francis D. Prevalence of virulence genes in *Escherichia coli* strains recently isolated from young pigs with diarrhea in the US. *Vet Microbiol* 2007;123:145–52. <https://doi.org/10.1016/j.VETMIC.2007.02.018>
- [33] Kim SJ, Jung WK, Hong J, Yang S-J, Park YH, Park KT. Prevalence and Characterization of Virulence Genes in *Escherichia coli* Isolated from Diarrheic Piglets in Korea. *J Food Hyg Saf* 2020;35:271–8. <https://doi.org/10.13103/JFHS.2020.35.3.271>
- [34] Kylla H, Dutta TK, Roychoudhury P, Subudhi PK, Lalhruaipuii, Lalsiamthara J, et al. Characterisation of Porcine Enteropathogenic *Escherichia coli* Isolated in Northeastern India. *J Vet Res* 2020;64:391. <https://doi.org/10.2478/JVETRES-2020-0046>
- [35] Rajkhowa S, Sarma DK. Prevalence and antimicrobial resistance of porcine O157 and non-O157 Shiga toxin-producing *Escherichia coli* from India. *Trop Anim Health Prod* 2014;46:931–7. <https://doi.org/10.1007/S11250-014-0587-4>
- [36] Kaufmann M, Zweifel C, Blanco M, Blanco JE, Blanco J, Beutin L, et al. *Escherichia coli* O157 and non-O157 Shiga toxin-producing *Escherichia coli* in fecal samples of finished pigs at slaughter in Switzerland. *J Food Prot* 2006;69:260–6. <https://doi.org/10.4315/0362-028X-69.2.260>
- [37] von Mülling T, Smajilovic M, Nowak B, Sammet K, Bülte M, Klein G. Preliminary study of certain serotypes, genetic and antimicrobial resistance profiles of verotoxigenic *Escherichia coli* (VTEC) isolated in Bosnia and Germany from cattle or pigs and their products. *Int J Food Microbiol* 2007;117:185–91. <https://doi.org/10.1016/j.jfoodmicro.2006.08.003>
- [38] Moredo FA, Cappuccio JA, Insarralde L, Perfumo CJ, Quiroga MA, Leotta GA. Genotypic characterization of toxigenic *Escherichia coli* isolated from pigs with postweaning diarrhea (PWD) and edema disease (ED). *Rev Argent Microbiol* 2012;44:85–8.
- [39] Mohlatlole RP, Madoroba E, Muchadeyi FC, Chimonyo M, Kanengoni AT, Dzomba EF. Virulence profiles of enterotoxigenic, shiga toxin and enteroaggregative *Escherichia coli* in South African pigs. *Trop Anim Health Prod* 2013;6:1399–405. <https://doi.org/10.1007/S11250-013-0377-4>
- [40] Ng PC, Henikoff S. Predicting the effects of amino acid substitutions on protein function. *Annu Rev Genom Hum Genet* 2006;7:61–80. <https://doi.org/10.1146/annurev.genom.7.080505.115630>
- [41] Yang G-Y, Guo L, Su J-H, Zhu Y-H, Jiao L-G, Wang J-F. Frequency of Diarrheagenic Virulence Genes and Characteristics in *Escherichia coli* Isolates from Pigs with Diarrhea in China. *Microorganisms* 2019;7:308. <https://doi.org/10.3390/microorganisms7090308>
- [42] Palma E, Tilocca B, Roncada P. Antimicrobial resistance in veterinary medicine: an overview. *Int J Mol Sci* 2020;21:1914. <https://doi.org/10.3390/ijms21061914>
- [43] Olorunmola FO, Kolawole DO, Lamikanra A. Antibiotic resistance and virulence properties in *Escherichia coli* strains from cases of urinary tract infections. *Afr J Infect Dis* 2013;7(1):1–7. <https://doi.org/10.4314/ajid.v7i1.1>
- [44] Grakh K, Mittal D, Parkash A, Bangar YC. Assessing the potential risk factors associated with avian colibacillosis using a questionnaire survey. *Har Vet* 2020;59:71–4.
- [45] Gosling R. A review of cleaning and disinfection studies in farming environments. *Livestock* 2018;23:232–7. <https://doi.org/10.12968/live.2018.23.5.232>
- [46] Kilungo A, Powers L, Arnold N, Whelan K, Paterson K, Young D. Evaluation of well designs to improve access to safe and clean water in Rural Tanzania. *Int J Environ Res Public Health* 2018;15:64. <https://doi.org/10.3390/ijerph15010064>